Amrita Vishwa Vidyapeetham, Kochi Campus

School of Computing

End Semester Examination, May 2024

2023 Integrated MCA/BCA II Semester

21CSA112- Database Management System

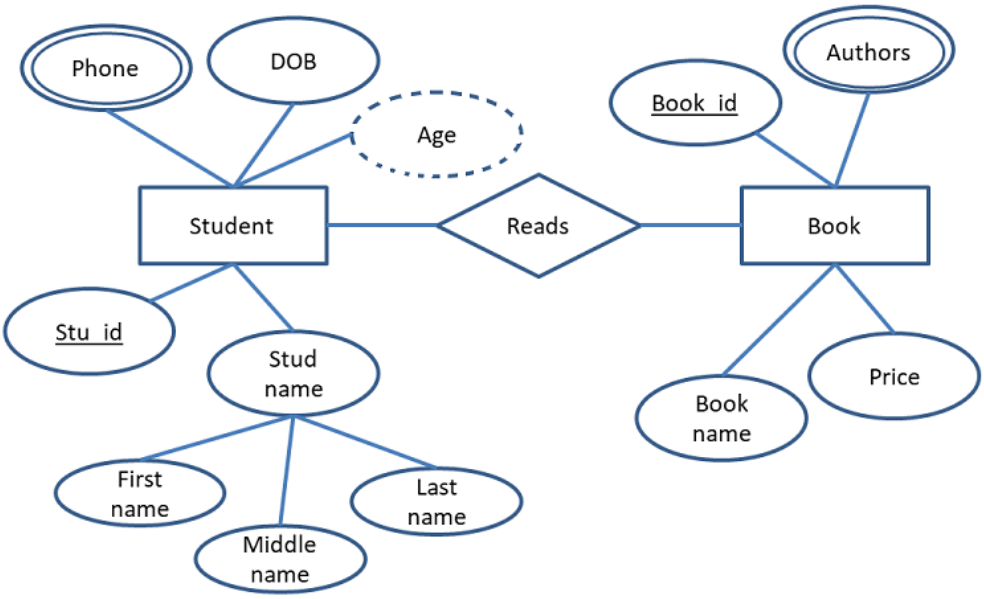
Answer all Questions

Time: 3 Hours **Maximum Marks : 100**

PART-A

(10x2=20)

1. Differentiate between physical data independence and logical data independence. Which independence is difficult to obtain and why?
2. State and define the following rules in Codd's Rule
   a. The Information Rule
   b. The Guaranteed Access Rule
3. Identify the following components in the given below ER-Diagram



   a. Derived attributes
   b. Multi-valued attributes
   c. Composite attributes

d. Primary keys

4. Define entity integrity constraints with examples.

5. State the rule of functional dependency on Relation R(A,B) A->B. Identify which among the following functional dependencies exists in the below table
   a. STUD_NO->STUD_NAME
   b. STUD_NO->STUD_PHONE
   c. STUD_NAME->STUD_STATE
   d. STUD_NAME->STUD_AGE

STUDENT

STUD_NO	STUD_NAME	STUD_PHONE	STUD_STATE	STUD_COUNTRY	STUD_AGE
1	RAM	9716271721	Haryana	India	20
2	RAM	9898291281	Punjab	India	19
3	SUJIT	7898291981	Rajsthan	India	18
4	SURESH		Punjab	India	21

Table 1

6. State the given below Armstrong's Axioms with example
   a. Reflexivity Axiom:
   b. Augmentation Axiom:
7. Define ACID test with reference to transactions. List out the commands that is used to manage transaction in SQL.
8. Illustrate the situation of deadlock in DBMS with example. How can we recover from a deadlock?
9. List out the attributes of a user defined cursor .
10. Debug the given below PL/SQL code block

```
Declare
Eno Emp.Empno %rowtype=&eno;
Salary Emp.Sal %rowtype;
Begin
Select Sal from emp into Salary where empno=Eno;
Dbms.output.put_line("SaLary ="| Sal);
Exception
When no_data_found
Dbms.output.put_line("No such Employee");
End;
```

PART-B (5x10=50)

11. Illustrate the three level architecture of database system with the help of a neat diagram
12. Consider the following Tables:

    Employee (emplD, FirstName, LastName, address, DOB, sex, position, deptNo, Salary)
    Department (dtptNo, deptName, mgr, empID)
    Project (projNo, projName, deptNo)
    Work on (empID, projNo, hours worked)

    Write the SQL statements for the following:

    a) Command to create the tables
    b) Insert a record inside Employee table
    c) Update the salary of all employees in dept 10 with a 5%increase
    d) Add a column (Location) in department table
    e) Get the names of employees along with their dept names.
    f) Get the names of employees along with number of projects assigned to each employee.
    g) Get the employee name and total number of hours worked by each employee in 'Inventory' project.
    h) Using sub-query find the names of employees in 'Sales' Department
    i) Delete the details of employees whose salary is <20000
    j) Remove the table Project

13. Define the following terms with example.
    a. Table level and column level constraints
    b. Group by and having
    c. Join
    d. Set operations
    e. Sub-Query

14. Define Transaction in DBMS. Illustrate the possible states though with a transaction may pass though during its life cycle with the help of a diagram.
15. Define concurrency control in DBMS. List out the problems that are possible to arise while executing concurrent transactions.

PART-C
(2x15=30)

16. Consider the following University information System. The university has different courses and conducts exams for students. The grade of a student is based on the Avg_Range of marks obtained in each exam.
    a. Draw an ER-Diagram for the above system and convert it to relational (6 marks)
    b. Normalize the below University Database (1NF, 2 NF, 3NF, BCNF). Find out different types of anomalies that occur in 1NF and 2NF) (9 marks)

Course_id	Course_name	Student_id	Student_name	Exam_name	Avg_Range	Grade
C1	BCA	{B1,B2}	{JHONES,JAMES}	{P1,P2}	{70-80,80-90}	{A2,A1}
C2	MCA	{M1,M2}	{HARRY,ANN}	{P2,END_SEM	{50-60,80-90}	{B1,A1}

17. Consider the following Tables

    {item_master-Item_code, Item_name, Price, stock}

    {Sales-Item_code,Qty_saled,Total_price}

a) Write a PL/SQL block to insert into Sales Table. While Inserting check whether item exists in item_master table. If exists update the stock of the item_master table with Stock-Qty_saled. Also check for in-sufficient stock. Display error if item does not exist in the master table. (9 marks)

b) Write a PL/SQL block using cursor to display the details of all items whose stock is below 100. (6 marks)